

GPT-5.4 Thinking

Step 1: Assess the hypothesis clarity and significance. The hypothesis is clear, specific, and meaningful. It makes a concrete claim: activation editing can be combined with differential privacy by perturbing steering vectors with calibrated Gaussian noise, yielding a method that preserves alignment behavior while providing formal privacy guarantees and minimally harming general model capabilities. This is not trivial; it targets an important and underexplored intersection of LLM alignment and privacy.

Step 2: Assess whether the experiments are aligned with the hypothesis. The experiments map well onto the core claims: - alignment effectiveness is tested in Experiment 1, - open-ended behavior quality is tested in Experiment 2, - preservation of general capabilities is tested in Experiment 3, - scaling and privacy-utility tradeoff are tested in Experiment 4, - privacy leakage and empirical auditing are tested in Experiment 5. This is a strong sign of rigor because the empirical plan covers the major components of the hypothesis rather than only a narrow slice.

Step 3: Evaluate baselines and controls. The proposal includes relevant non-private activation-editing baselines (Mean Steering, PCA Steering) and zero-shot unsteered baselines. These are appropriate comparisons for showing the cost of privacy and the value of steering. The inclusion of both private and non-private methods is important for isolating the effect of DP noise. However, the baseline set is somewhat narrow: there is no comparison against stronger alignment methods such as fine-tuning-based private methods, RLHF-style alternatives, or other privacy-preserving adaptation techniques if such exist. This weakens completeness somewhat, but does not fundamentally undermine the design.

Step 4: Evaluate the metrics. Some metrics are appropriate and standard: - MCQ accuracy for alignment benchmark choices, - MMLU performance for general capability retention, - privacy attack metrics such as FPR/FNR. But there are also notable weaknesses: - Experiment 2 relies on GPT-4 qualitative evaluation, which can be useful but is subjective and vulnerable to evaluator bias unless carefully rubricized and validated. - Experiment 3 uses average probability on the correct answer for MMLU instead of the more standard accuracy; this is not necessarily wrong, but it is less interpretable and may complicate comparisons. - The reported privacy metric 'empirical epsilon' from an attack is not equivalent to the formal DP guarantee and could be misleading if not clearly separated from theoretical privacy accounting. Overall, the metrics are mostly sensible but not uniformly strong.

Step 5: Evaluate methodological rigor of the privacy claim. This is the most important issue. The hypothesis claims formal privacy guarantees via DP, and the experiment description says Gaussian noise is calibrated and privacy leakage will be audited empirically. This suggests awareness of both theoretical and empirical privacy evaluation. However, the experiment description does not specify crucial DP methodology details such as: - the neighboring dataset definition, - sensitivity analysis of steering vector computation, - privacy accounting across tasks/models/layers, - whether hyperparameter tuning consumes privacy budget, - composition across repeated evaluations, - clipping or boundedness assumptions required for Gaussian mechanisms. Without these details, the experimental plan supports the idea of evaluating a DP method, but does not fully demonstrate that the formal privacy claim is implemented rigorously. This is a significant concern, though the presence of both formal-DP intent and attack-based auditing is better than many weak proposals.

Step 6: Evaluate breadth and robustness. The plan is fairly broad: - multiple model families and sizes, - multiple alignment dimensions, - both closed-form MCQ and open-ended evaluations, - general capability tests, - scaling studies, - adversarial privacy audits. This breadth is a strong marker of rigor. The scaling study is especially helpful because it tests whether the proposed mechanism behaves systematically rather than only on a single model.

Step 7: Identify missing ablations or controls. Important ablations are not explicitly described: - varying privacy levels/noise scales and plotting privacy-utility curves more systematically, - layer-wise steering choices, - number of private demonstrations, - effect of clipping or vector normalization, - sensitivity to random seeds, - comparison of different attack strengths in the privacy audit. These omissions reduce rigor. In particular, for a DP method, one would expect explicit sweeps over epsilon/delta and confidence intervals across seeds.

Step 8: Judge whether the flaws are fundamental or moderate. The flaws are moderate rather than fundamental. The proposal has a strong central hypothesis, relevant baselines, multi-angle evaluation, and a privacy-specific attack audit. The main weaknesses are incomplete specification of the DP accounting and some evaluation choices that are weaker than ideal. Still, the overall design is much stronger than a low-rigor submission, which would typically have vague claims, missing baselines, or no real privacy evaluation.

Step 9: Final bucket decision. Despite some methodological gaps, **the proposal demonstrates a clear and meaningful hypothesis and a generally rigorous experimental plan with appropriate controls, multiple evaluations, and explicit attention to privacy-utility tradeoffs and leakage auditing. Therefore it fits the high rigor bucket better than low.**

Step 10: Confidence assessment. I am confident but not absolutely certain. The classification depends partly on assuming the formal DP mechanism is implemented correctly, while the provided description omits some technical details needed to verify that. So confidence is 4 rather than 5.